

A new life

THE TIME IN THE WOMB IS THE FIRST, yet largely unseen, part of a baby's life. It is the period between an egg joining with a sperm cell at fertilization and the resulting baby's entrance into the world at birth. A female human is pregnant for about 266 days. A female mouse is pregnant for about 20 days and a female elephant for some 660 days. Our species follows the general trend in the mammal group – the larger the body size, the longer the pregnancy. In fact, almost everything about pregnancy in humans follows the standard mammal pattern. Knowledge of pregnancy has come a long way since Aristotle's views. He stated that the father provided the soul and spirit for the baby. The mother provided the menstrual blood, which mixed with semen fluid from the father and remained in the womb to give rise to the baby. The mother was thought to be simply a carrier, or incubator, feeding and providing for the life force that originated from the man.



FIRST EMBRYOLOGIST
In 1604, Italian professor Hieronymus Fabricius (1537-1619) published *De Formato Foetu*, a study about the development of eggs and unborn babies in a range of animals, including humans. Even in his lifetime, Fabricius was known as the founder of embryology. He named the ovary, the egg-making organ in the chicken, and predicted its function.



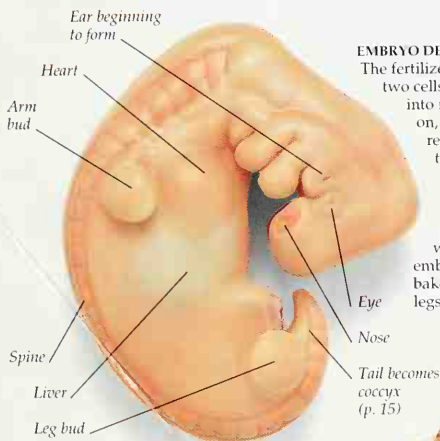
SEXUAL INTERCOURSE

In the 1950s and 1960s, scientists carried out detailed research into sexual intercourse. The woman's vagina (p. 43) expands and moistens with mucous secretions. The man's penis (p. 42) fills with blood and becomes erect. In the orgasmic phase of the male, the muscles around the sex organs contract rhythmically, and sperm cells in their semen fluid are ejaculated into the woman's vagina. After orgasm, the muscles and sex organs relax and return to normal, and the partners sometimes feel drowsy.



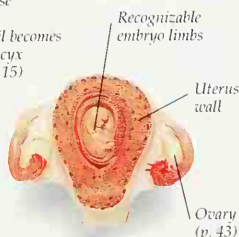
FERTILIZATION OF THE EGG CELL

In this electron micrograph, a sperm cell is penetrating the follicular cells around the egg cell (p. 43). The sperm cell is about 0.002 in (0.05 mm) long, and most of this is its long tail, trailing off toward the top left in the picture. There are about 300-500 million sperm in the semen fluid, but only one sperm can fertilize the egg.



EMBRYO DEVELOPMENT

The fertilized egg divides into two cells (pp. 48-49), then into four, then eight, and so on, every few hours. The resulting hundreds and then thousands of cells gradually group together, and change into different types, such as muscle, nerve, and blood cells. Five weeks after fertilization, the embryo is smaller than a baked bean, yet its arms and legs are already developing.

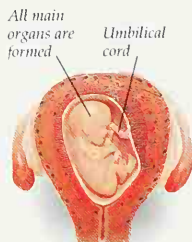


FETAL DEVELOPMENT

About two months after fertilization, all of the major organs have formed, the baby's heart is beating, and yet its body is still just 1 in (25 mm) long. Embryologists through the ages have been amazed at the similarity between the early embryos of humans, monkeys, cats, and many other types of mammals. From two months through to birth, the developing baby gradually comes to look distinctly human. It is known as a fetus, from the Latin for "offspring."

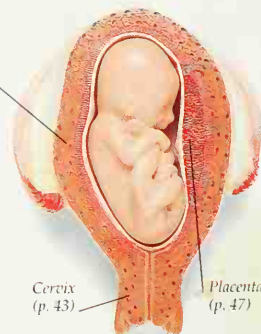
1 TWO MONTHS

The baby is smaller than a walnut, and tiny fingers and toes are growing. The pregnancy is confirmed.



2 THREE MONTHS

About 2.8 in (70 mm) long, the baby can move its head and limbs. The mother hardly feels this.



3 FIVE MONTHS

The baby is 10 in (25 cm) long and responds to loud sounds by kicking and turning somersaults. The mother's abdomen bulges.